

Weekend Assessment – 1

Section 1: Python Foundations (20 Minutes)

1. **Filtering Data:** You have a list of sensor readings: `readings = [25.1, 18.5, 32.0, 15.2, 28.9]`. Write a for loop that prints only the readings that are greater than **20**.
2. **Basic Arithmetic (The Mean):** Write a script to calculate the average of the following list: `scores = [80, 90, 70, 100]`. Do not use any external libraries.
3. **String Formatting:** Given the variables `accuracy = 0.95` and `model_name = "Neural Network"`, print the following sentence using an f-string: *"The Neural Network model has an accuracy of 95%."*
4. **List Operations:** Create a list called `features` with the items "height" and "weight". Add the item "age" to the end of the list, then remove "height".
5. **Boolean Logic:** In ML, we often check if a value is within a range. Write an if statement that checks if a variable `x` is between 0 and 1 (inclusive) and prints "Valid" if it is.

Code:

1.

```
readings = [25.1, 18.5, 32.0, 15.2, 28.9]
for r in readings:
    if r > 20:
        print(r)
```

2.

```
scores = [80, 90, 70, 100]
mean = sum(scores) / len(scores)
print(mean)
```

3.

```
accuracy = 0.95
model_name = "Neural Network"
print(f"The {model_name} model has an accuracy of {int(accuracy * 100)}%.")
```

4.

```
features = ["height", "weight"]
features.append("age")
features.remove("height")
print(features)
```

5.

```
x = 0.5
if 0 <= x <= 1:
    print("Valid")
```

Section 2: Intro to Data Arrays (20 Minutes)

6. **Importing Libraries:** Write the standard line of code used to import the **NumPy** library so that you can refer to it as **np**.
7. **Creating Arrays:** Using NumPy, create an array containing the numbers **1 through 5**.
8. **Array Shapes:** If you have a NumPy array called **data** with 10 rows and 3 columns, what code would you write to find and print the **shape** of that array?
9. **Simple Scaling:** You have a list **prices = [100, 200, 300]**. Use a list comprehension to create a new list where every price is divided by 100 (this is a basic form of scaling).
10. **Finding the Max:** Given the array **errors = [0.5, 0.1, 0.9, 0.2]**, use a built-in Python function to find and print the **largest** error value in the list.

Code:

6.

```
import numpy as np
```

7.

```
arr = np.array([1, 2, 3, 4, 5])
```

8.

```
print(data.shape)
```

9.

```
prices = [100, 200, 300]
scaled = [p / 100 for p in prices]
print(scaled)
```

10.

```
errors = [0.5, 0.1, 0.9, 0.2]
print(max(errors))
```